CIDM/ECON 6308 LA5 Time Series Decomposition

Objectives:

- Understand Time Series Decomposition by explaining the purpose of decomposing a time series into trend-cycle, seasonal, and remainder components;
- Apply Time Series Decomposition by executing decomposition procedures on provided datasets and interpret the results in Excel or RM;
- Analyze Time Series Components by identifying the number of observations available for each decomposed component;
- Develop Data Interpretation Skills by relating decomposition results to economic and business trends in retail trade.

Instructions

In the subfolder "Class 11 Lab & LA5" under Class 11 folder, the instructor provided two examples for time series decomposition: the first example/video demonstrates how to decompose the soft drink data in Excel, and the second one demonstrates how to decompose the beers data in RM. Please follow the instructions in the videos to complete the practice tasks.

After completing the two practice tasks, please follow similar procedures to decompose the following quarterly time series: Net Sales, Receipts, and Operating Revenues of all Retail trade in the United States from Q1 in 2001 to Q4 in 2024 (in Millions of Dollars). Please download the dataset "Class 11 Lab Dataset" on WTCLASS for this lab, instead of the dataset on the FRED website.

You can choose Excel or RM to decompose the time series using the classical additive method and then answer the following questions:

1. In total, the estimate of the trend-cycle component is available for 92 observations; the estimate of the seasonal component is available for 96 observations; the estimate of the remainder component is available for 92 observations; the estimate of the seasonally adjusted series is available for 96 observations (type a whole number for each blank).

-22778 Excel

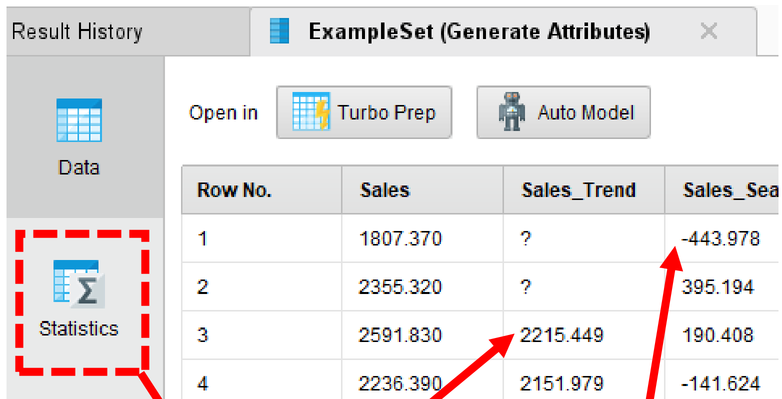
2. The estimate of the first available trend-cycle component is 341098 million dollars; the estimate of the first available seasonal component -22630 million dollars; the estimate of the first available remainder component is -2325 million dollars; the estimate of the first available seasonally adjusted value is 340644 million dollars. (Round each answer to a whole number).

-2348 Excel

340792 Excel

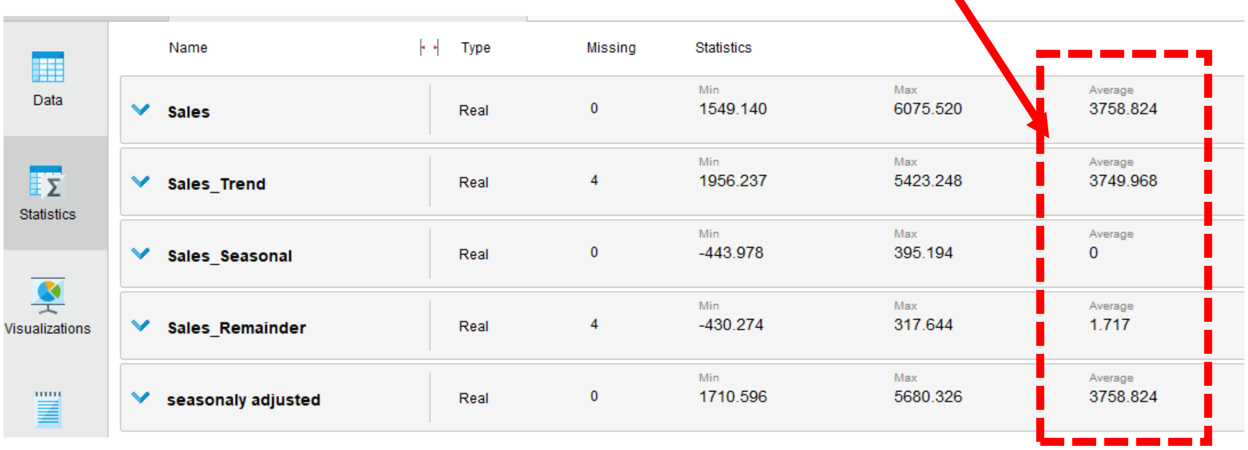
3. Please find or compute the average for each series. After decomposing the series, you can find those averages under the Statistics view in RM (Please refer to the screenshots of different decomposed time series below) or use a function in Excel to find those averages. Round each answer to a nearest whole number and DONOT include units in your answers.

- The average of revenues is 646211 million dollars.
- The average of the trend-cycle component is 643931 million dollars. 554387 Excel
- The average of the seasonal component is 0 million dollars.
- The average of the remainder component is -113 million dollars.
- The average of the seasonally adjusted series is 646211 million dollars.



This is the estimate of the first available trend-cycle component.

This is the estimate of the first available seasonal component.



4. Based on the time plot or the decomposition result, generally, Quarter 4 has the highest revenue, while Quarter 1 has the lowest revenue. Type 1, 2, 3, or 4 here.